SUBHROJYOTI SARKAR

Kharagpur, West Bengal | subhrojyotisarkar60@gmail.com |https://www.linkedin.com/in/subhrojyoti-sarkar/

EDUCATION

VELLORE INSTITUTE OF TECHNOLOGY — Amaravathi, Andhra Pradesh
Bachelor of Technology — Sept 2023 – June 2027
Major in Computer Science
Cumulative GPA: 8.61/10 (till 7th semester)

DAV PUBLIC SCHOOL — Jamshedpur, Jharkhand
CBSE — April 2020 – May 2022
Percentage: 79 %

KENDRIYA VIDYALAYA NO.2, A.F.S — Air Force Station, Kalaikunda
CBSE — April 2010 – July 2020
Percentage: 91.8 %

PROJECTS

HYBRID-SEARCH RAG with CITATION VERIFICATION | ***Python, FastAPI, FAISS, BM25, Sentence Transformers, Gemini API, Docker***

- Built a PDF question-answering system that combines **BM25** and **FAISS dense retrieval** via Reciprocal Rank Fusion, with **cross-encoder reranking** and cites every claim to its exact document, page and passage.
- Designed an **independent LLM-citation verifier** whose evidence quotes are validated in code; unsupported claims are removed, and the system abstains when the documents lack the answer.
- Created a **human-labelled evaluation framework** that guided retrieval fixes such as table-aware chunking and a corrected BM25 IDF; the system ships with role-based login, Docker deployment and **268 automated tests**.

AI Resume Screener and Job Matcher | ***React, Node.js, Python (FastAPI), scikit-learn, PostgreSQL, MongoDB, Docker***

- Built a full-stack app with **React and Node.js/Express** where recruiters post jobs, upload resumes, and view candidates ranked by match score.
- Developed a **FastAPI** service using **sentence-embeddings,** TF-IDF and skill matching (**PostgreSQL, MongoDB**), containerized with **Docker Compose** and backed by JWT auth and **150+ automated tests**.
- Automated resume screening into **ranked, explainable shortlists** (matched and missing skills per candidate), using **semantic matching** to surface relevant candidates that keyword search scored low.

SKIMLIT – RESEARCH PAPER CLASSIFIER

- Built an **NLP-based AI tool** that classifies medical abstract into **Objective, Methods, Results and Conclusion** for improved readability.
- Implemented **TensorFlow and Transformer-based architecture (BERT)** for accurate text classification and summarization.

CERTIFICATION

- Building Agents with Agentic AI (FDPJf4doou, provided by IBMCE) ID: **3cc5ff9079cb45e98fa61a5eb5282d17**

LEARDERSHIP

ENTREPRENEURSHIP CLUB — Aug 2024 – June 2025

- Served as Joint Secretary, leading initiatives to support campus startups and entrepreneurial activities.
- Guided the team to secure 17th in All-India rank in the National Entrepreneurship Challenge 2025 at IIT Bombay.

SKILLS

- Languages**: Java, Python, HTML, CSS, SQL, JavaScript**
- Libraries/Frameworks**: FastAPI, React, Node.js, Express.js, Tensorflow, scikit-learn**
- Databases**: PostgreSQL, MongoDB**
- Tools & Cloud**: Docker, Docker Compose, Git, Jupyter Notebook, Pandas, NumPy**